



DEIVA MANI KAILASAM

Senior Software Engineer (Full Stack, AI)

Cloud-Native Distributed Systems | GenAI & Agentic AI Applications

✉ deivamanikailasam@gmail.com ☎ +61 411 181 052

🌐 [LinkedIn: deivamani-k] 📁 [Portfolio]

SKILLS

- **Languages & Application Stack:** Python, TypeScript, Java, Spring Boot, Node.js/Express, JavaScript, React, Angular, Next.js, C#
- **GenAI & Agentic AI:** LLMs, RAG pipelines, embeddings, prompt engineering, Vector DBs, MCP tools & integrations, LangChain, LangGraph, LlamaIndex, Snowflake Cortex, LLM-powered analytics, multi-agent orchestration, AI-assisted developer workflows, Claude, OpenAI, Cursor AI, Github Copilot
- **Frontend & UI Engineering:** Tailwind CSS, SCSS, PrimeNG, Angular Material, AG Grid Enterprise, Highcharts, Storybook, Compodoc, MDX, white-label theming, design systems
- **APIs, Data & Messaging:** FastAPI, REST, gRPC, GraphQL, OpenAPI/Swagger, PostgreSQL, SQL Server, Snowflake, MongoDB, MongoDB Atlas Search, Kafka, Redis, Caffeine, JSONB, full-text search, Liquibase
- **Cloud, DevOps & Platform:** AWS (ECS, Fargate, EC2, Lambda, EKS, Bedrock, SageMaker), Azure (App Service, Functions, PostgreSQL, SQL, Cosmos DB, Blob Storage, AD B2C, Azure DevOps), Docker, Kubernetes, Terraform, Helm, ArgoCD, GitHub, Jenkins, ARM/Bicep

PROFILE

Senior Software Engineer (Full Stack, AI) with 9 years of experience building GenAI-enabled, cloud-native, multi-tenant SaaS platforms on AWS and Azure using Python, TypeScript, Node.js, Java/Spring Boot, React, Angular, and Next.js. Delivered production systems combining intelligent backend services, interactive data-rich frontends, and strong platform engineering foundations, including distributed APIs handling 10K+ daily requests and products serving 100K+ users. Specialized in GenAI and agentic applications using RAG, tenant-scoped embeddings, LLM-powered analytics, LangChain, LangGraph, LlamaIndex, and Snowflake Cortex, deployed to production via AWS services. Brings end-to-end ownership across business process, architecture, implementation, testing, CI/CD, observability, reliability, governance and mentoring.

WORK EXPERIENCE

Kinesso, Omnicom Media Group

AUG 2021 - JULY 2026

Senior Software Engineer (Full Stack, AI)

GenAI & Agentic Platforms

- Worked with business and product stakeholders to qualify GenAI use cases, define success metrics and automation levels, and build ROI cases before committing to build. Ran feasibility spikes on real data to de-risk GenAI projects and support go/no-go decisions.
- Designed and implemented scalable Python FastAPI services and data pipelines for multi-tenant GenAI applications, integrating RAG pipelines with tenant-specific embeddings stored in MongoDB Atlas vector search and handling 10K+ daily inference requests with sub-200ms latency.
- Designed GenAI pipelines for tenant-aware embeddings, LLM-based anomaly detection, and a Python-based data quality scoring engine, reducing undetected data quality issues around 75%.
- Developed agentic AI prototypes using LangChain and LangGraph for multi-agent orchestration and document-querying workflows integrated with production data systems, partnering with marketing and analytics stakeholders to validate outputs against business KPIs.
- Built a Snowflake analytics agent using LangGraph and Snowflake Cortex for natural-language querying and dynamic dashboard generation, enabling business users to explore data and surface analytics insights through interactive dashboards.
- Deployed small open-weight models (Llama 3) on Amazon Bedrock as intent routers for multi-agent workflows, with orchestration handled in LangGraph.
- Applied frontier models (GPT, Claude Sonnet/Opus) on Amazon Bedrock for reasoning, planning, and advanced tool use with prompt caching to balance quality, latency, and inference cost.
- Secured RAG and agent workflows with input/output guardrails — Bedrock Guardrails for PII/DLP and a fine-tuned small Llama classifier (via SageMaker) for prompt-injection and topical filtering — plus entitlement-aware retrieval to enforce tenant isolation.
- Built automated evaluation suites with versioned golden datasets and LLM-as-judge grading to measure GenAI quality and catch regressions before release.

- **Engineering Practices:** Test automation, Pytest, JUnit, Jest, Cypress, Playwright, MockMvc, clean code, code review, performance tuning, incident response, Datadog, Application Insights, Copilot, Cursor AI, Claude, Agile
- **Architecture & Design:** System design, microservices, distributed systems, multi-tenant SaaS, API design, CQRS, Saga, reactive programming, SQL/NoSQL data modelling, test-driven development

NOTABLE PROJECTS

- **Snowflake Agent:** Built a Python-based analytics agent using LangGraph and Snowflake Cortex for natural-language querying and dynamic dashboard generation over Snowflake data.
- **AI Agent for Code Generation:** Built an LLM-powered internal engineering tool using GPT-4 and LangChain to generate boilerplate and integration code, accelerating delivery for internal teams.
- **Feature-Rich UI Component Library:** Built a multi-framework design system across Angular, React, and Next.js with 40+ reusable components used across multiple internal products.
- **Edge Water Management Platform:** Delivered a full-stack web application for smart water monitoring and optimization using Angular and cloud backend services.

- Shipped agent features through shadow and canary rollouts with human-in-the-loop approvals, kill switches, and trace-based observability feeding a continuous feedback loop.

Backend, APIs & Data Platforms

- Architected and led a multi-module Spring Boot 3.x microservices platform with JWT/OAuth2, multi-tenant data access, and OpenAPI contracts, serving 10K+ daily API requests across PostgreSQL, SQL Server, Snowflake, and MongoDB.
- Designed multi-tenant isolation patterns using DB-per-tenant, collection-per-tenant, and row-level security, ensuring zero cross-tenant leakage across 12+ environments.
- Implemented a gRPC-based plugin architecture for integrating Python microservices and client-specific React dashboard plugins, improving onboarding speed for new services.
- Built a hybrid data access layer spanning PostgreSQL, SQL Server, Snowflake, and MongoDB with Liquibase migrations and MongoDB schema versioning, simplifying cross-database querying.
- Integrated Google Ads, Meta/Facebook, Bing Ads, Adverity, Snowflake, Power BI, and MongoDB Atlas Search using OAuth2/OIDC and Python data connectors, enabling near real-time advertising analytics.
- Reduced 99th percentile latency by about 55% through Spring WebClient optimization, Python asyncio, MongoDB GridFS, parallel query execution, and hybrid caching using Caffeine, Redis, and MongoDB.

Frontend & Product Experience

- Built and maintained a multi-framework design system with 40+ reusable components across React, Angular, and Next.js, distributed through private npm and adopted across 6 internal products.
- Implemented advanced AG Grid Enterprise tables with filtering, sorting, row expansion, and CSV/Excel export through ExcelJS, enabling non-technical users to self-serve analytics.
- Delivered a white-label theming system across React, Next.js, and Angular supporting 5+ brand variants with dynamic stylesheet injection.
- Authored Storybook documentation with 150+ stories, accessibility tests, and AI-assisted component suggestions, reducing duplicate components.

Platform, Quality & Leadership

- Established automated testing across 1000+ unit tests and 100+ integration/E2E tests using Jest, Vitest, Pytest, Cypress, Playwright, and MockMvc, achieving 80%+ coverage across Java, Python, and UI layers.
- Configured CI/CD with Jenkins and GitHub Actions for React/Next.js using multi-stage Docker builds, SonarQube gates, and pull-request validation, cutting merge-to-deploy time.
- Designed Kubernetes infrastructure for Spring Boot, Python FastAPI, and Next.js workloads using ConfigMaps, Vault secrets, MongoDB StatefulSets, health probes, and Datadog dashboards, reducing MTTR.
- Migrated CI/CD pipelines to hybrid AWS environments using Terraform and ArgoCD, automating Docker and Kubernetes deployments of Python AI services with Helm charts, Kyverno policy-as-code, and reusable GenAI library modules.
- Developed Terraform infrastructure for AWS ECS/Fargate and MongoDB Atlas with GitOps via ArgoCD, enabling one-command deployments and eliminating manual infrastructure changes.
- Mentored 5+ engineers, with 3 promoted to mid-level roles, and drove coding standards across Python, TypeScript, Java, and multi-service codebases.

ACE Contractors Pvt Ltd

Software Developer (Full Stack)

DEC 2020 - JUL 2021

- Architected and owned a Progressive Web App platform for construction operations on Azure App Service, Azure PostgreSQL Flexible Server, Azure Functions, Blob Storage, and AD B2C, serving 200+ daily active users across 10+ sites.
- Built a mobile-first Angular frontend with PWA capabilities including Service Workers, App Manifest, and Cache API, enabling offline workflows for field workers and improving daily task completion rates by 40%.

EDUCATION

- **Master of Technology in Information Technology**

La Trobe University

Jul 2019 – May 2021

- **Bachelor of Technology in Information Technology**

Valliammai Engineering College

Aug 2011 – May 2015

CERTIFICATIONS

- Google Cloud Certification on **GenAI** (2025)
- IBM Certification on **Prompt Engineering** (2025)
- **Best IT Project** Award – La Trobe University (2020)
- **Pursuit of Knowledge** Award, **On the Spot** Award, **Cornerstone** Award, **Star** Award – TCS (2016–2018)

- Implemented Node.js/Express backend services with gRPC microservices and stream processing for real-time timesheet validation and contract updates, achieving sub-100ms response times for 95th percentile requests.
- Designed PostgreSQL schemas for employees, projects, contracts, timesheets, and safety incidents using JSONB and full-text search for efficient retrieval.
- Delivered CI/CD using Azure DevOps YAML pipelines, multi-stage Docker builds, Azure Container Registry, and ARM/Bicep templates, reducing deployment time from 2 hours to 12 minutes.
- Led weekly executive syncs with the company owner, prioritizing high-ROI features that reduced payroll disputes, digitized timesheets, and accelerated management documentation and reporting workflows.

Tata Consultancy Services

JAN 2016 – MAY 2019

Software Developer (Full Stack) / Systems Engineer

- Led migration of a legacy Java EE monolith to Spring MVC and Spring Boot layered architecture, reducing technical debt and removing critical security vulnerabilities.
- Gradually migrated AngularJS 1.5 to Angular 5 using RxJS, Angular Router, Angular Material, and lazy loading, achieving 3x faster initial page loads and 65% bundle size reduction.
- Designed PostgreSQL schemas replacing Oracle for projects, employees, timesheets, and billing with stored procedures and indexing, cutting query times by 65% and improving page load performance.
- Implemented test-first development with JUnit, Mockito, TestNG, and Selenium WebDriver E2E testing, achieving 85% unit test coverage and catching regressions early.
- Containerized Spring Boot services using Docker multi-stage builds, enabling 8+ major releases that consolidated multiple legacy systems.